CHAPTER 1

VECTOR SPACES AND LINEAR EQUATIONS

1 Basic Structures and Definitions

Definition 1.1: Ring R

A set R with two operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ is called a ring if the following hold:

1. (Additive abelian group) $(R, +)$ is an abelian group with identity 0.
2. (Distributivity) $x(a+b) = x \cdot a + x \cdot b$ and $(x+y)a = x \cdot a + y \cdot a$

Definition 1.2: Field

$(\mathbb{F}, +, \cdot)$ is a field if $\mathbb{F}$ is a set with operations $+ : \mathbb{F} \times \mathbb{F} \to \mathbb{F}$ and $\cdot : \mathbb{F} \times \mathbb{F} \to \mathbb{F}$ such that the following hold:

1. $(\mathbb{F}, +)$ is an abelian group with an identity element 0, i.e. $\forall a, b, c \in \mathbb{F}$, we have $a + (b + c) = (a + b) + c$ (Associativity), $a + b = b + a$ (Commutativity), $\exists 0 \in \mathbb{F}$ such that $a + 0 = a$, and $\forall a \in \mathbb{F}$, $\exists -a \in \mathbb{F}$ such that $a + (-a) = 0$.
2. $(\mathbb{F} \backslash \{0\}, \cdot)$ is also an abelian group, i.e. $\forall a, b, c \in \mathbb{F} \backslash \{0\}$, we have $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (Associativity), $a \cdot b = b \cdot a$, $\exists 1$ such that $a \cdot 1 = a$ and $\forall a \in \mathbb{F} \setminus \{0\}$ there exists a unique a^{-1} such that $aa^{-1} = 1$.
3. (Distributivity) $\forall a, b, c \in \mathbb{F}$, $a(b + c) = ab + ac$.

Such a structure is called a *Field*.